

|                                                     |                                                 |                                                 |                                                     |                                           |                                           |                                               |
|-----------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-----------------------------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                     | $\mathcal{K}_{0^+}^{\#1}$                       | $\mathcal{K}_{0^+}^{\#1}{}^{\alpha\beta\chi}$   |                                                     |                                           |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1}{}^\dagger$                 | 0                                               | 0                                               |                                                     |                                           |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1}{}^{\alpha\beta\chi}$       | 0                                               | $3k^2 \frac{(4)}{K_{15}} + 3 \frac{(2)}{K_1}$   | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$           | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$ | $Y_1 k^2 + 2 \frac{(2)}{K_1}$                   | $\frac{Y_2 k^2 + 4 \frac{(2)}{K_1}}{2\sqrt{2}}$ | 0                                                   | 0                                         |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$ | $\frac{Y_2 k^2 + 4 \frac{(2)}{K_1}}{2\sqrt{2}}$ | $Y_3 k^2 + \frac{(2)}{K_1}$                     | 0                                                   | 0                                         |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$      | 0                                               | 0                                               | $k^2 \frac{(4)}{K_1}$                               | $-\frac{Y_4 k^2}{2\sqrt{2}}$              |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$      | 0                                               | 0                                               | $-\frac{Y_4 k^2}{2\sqrt{2}}$                        | $\frac{Y_5 k^2}{2}$                       | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                     |                                                 |                                                 | $\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$ | 0                                         | 0                                         |                                               |
|                                                     |                                                 |                                                 | $\mathcal{K}_{2^+}^{\#1}{}^{\alpha\beta\chi}$       | 0                                         | 0                                         |                                               |

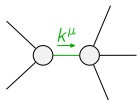
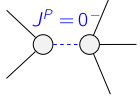
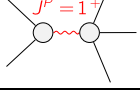
|                                                     |                                                                  |                                                                  |                                                     |                                               |                                           |                                               |
|-----------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                     | $\mathcal{J}_{0^+}^{\#1}$                                        | $\mathcal{J}_{0^+}^{\#1}{}^{\alpha\beta\chi}$                    |                                                     |                                               |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1}{}^\dagger$                 | 0                                                                | 0                                                                |                                                     |                                               |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1}{}^{\alpha\beta\chi}$       | 0                                                                | $\frac{1}{3(k^2 \frac{(4)}{K_{15}} + \frac{(2)}{K_1})}$          | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$           | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$     | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$ | $\frac{Y_3 k^2 + \frac{(2)}{K_1}}{\text{Det}(1^+)}$              | $\frac{-Y_2 k^2 - 4 \frac{(2)}{K_1}}{2\sqrt{2} \text{Det}(1^+)}$ | 0                                                   | 0                                             |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$ | $\frac{-Y_2 k^2 - 4 \frac{(2)}{K_1}}{2\sqrt{2} \text{Det}(1^+)}$ | $\frac{Y_1 k^2 + 2 \frac{(2)}{K_1}}{\text{Det}(1^+)}$            | 0                                                   | 0                                             |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$      | 0                                                                | 0                                                                | $\frac{Y_5 k^2}{2 \text{Det}(1^+)}$                 | $\frac{Y_4 k^2}{2\sqrt{2} \text{Det}(1^+)}$   |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$      | 0                                                                | 0                                                                | $\frac{Y_4 k^2}{2\sqrt{2} \text{Det}(1^+)}$         | $\frac{k^2 \frac{(4)}{K_1}}{\text{Det}(1^+)}$ | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                     |                                                                  |                                                                  | $\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$ | 0                                             | 0                                         |                                               |
|                                                     |                                                                  |                                                                  | $\mathcal{J}_{2^+}^{\#1}{}^{\alpha\beta\chi}$       | 0                                             | 0                                         |                                               |

#### Abbreviations used in matrices

$$\begin{aligned}
Y_1 &= 2 \frac{(4)}{K_{15}} - \frac{(4)}{K_3} \&\& Y_2 = 4 \frac{(4)}{K_{15}} + \frac{(4)}{K_4} \&\& \\
Y_3 &= \frac{(4)}{K_{15}} + \frac{(4)}{K_9} \&\& Y_4 = 2 \frac{(4)}{K_1} - \frac{(4)}{K_7} \&\& Y_5 = \frac{(4)}{K_1} - \frac{(4)}{K_3} - \frac{(4)}{K_4} - \frac{(4)}{K_7} + 2 \frac{(4)}{K_9} \&\& \\
\text{Det}(1^+) &= -\frac{1}{8} k^2 (k^2 (\frac{(4)}{K_4} + 8 \frac{(4)}{K_{15}} (\frac{(4)}{K_3} + \frac{(4)}{K_4} - 2 \frac{(4)}{K_9}) + 8 \frac{(4)}{K_3} \frac{(4)}{K_9}) + 8 (\frac{(4)}{K_3} + \frac{(4)}{K_4} - 2 \frac{(4)}{K_9}) \frac{(2)}{K_1})
\end{aligned}$$

#### Lagrangian

$$\begin{aligned}
&\frac{(2)}{K_1} \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \frac{(2)}{K_1} \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \frac{(4)}{K_1} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta - \\
&\frac{(4)}{K_1} \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta + \frac{(4)}{K_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \frac{(4)}{K_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \frac{(4)}{K_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \\
&\frac{(4)}{K_7} \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta \partial_\delta \mathcal{K}_{\chi\beta}^\delta + \frac{(4)}{K_9} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{(4)}{K_{15}} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \frac{(4)}{K_{15}} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

|                                                                                    |                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Source constraint(s)                                                               | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\mathcal{J}_{0^+}^{\#1} = 0$                                                      | 1                                                             | $\partial_\beta \mathcal{J}^\alpha{}_\alpha{}^\beta{}_\beta = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} = 0$                                       | 5                                                             | $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$<br>$4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\delta \mathcal{J}^{\delta\alpha}{}_\delta =$<br>$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$<br>$2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\delta \mathcal{J}^{\delta\beta}{}_\delta$ |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta} = 0$                                           | 5                                                             | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi}{}_\chi{}^\delta = 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi}{}_\chi{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Total # constraint(s):                                                             | 11                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Resolved pole(s)                                                                   | # polarization(s)                                             | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Residue                                                                                                                                                                                                                                                                                                                                                                                                                          |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $-\frac{\frac{(4)}{K_3}{}^2 + \frac{(4)}{K_4}{}^2 + \frac{(4)}{K_7}{}^2 + \frac{(4)}{K_4} (\frac{(4)}{K_7} - 4 \frac{(4)}{K_9}) + \frac{(4)}{K_3} (2 \frac{(4)}{K_4} + \frac{(4)}{K_7} - 4 \frac{(4)}{K_9}) - 2 \frac{(4)}{K_7} \frac{(4)}{K_9} + 4 \frac{(4)}{K_9}{}^2}{(\frac{(4)}{K_7} + 4 \frac{(4)}{K_1} (\frac{(4)}{K_3} + \frac{(4)}{K_4} - 2 \frac{(4)}{K_9})) (\frac{(4)}{K_3} + \frac{(4)}{K_4} - 2 \frac{(4)}{K_9})}$ |
|  | 1                                                             | $-\frac{\frac{(2)}{K_1}}{\frac{(4)}{K_{15}}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $-\frac{1}{3 \frac{(4)}{K_{15}}}$                                                                                                                                                                                                                                                                                                                                                                                                |
|  | 3                                                             | $-\frac{8 (\frac{(4)}{K_3} + \frac{(4)}{K_4} - 2 \frac{(4)}{K_9}) \frac{(2)}{K_1}}{\frac{(4)}{K_4}{}^2 + 8 \frac{(4)}{K_{15}} (\frac{(4)}{K_3} + \frac{(4)}{K_4} - 2 \frac{(4)}{K_9}) + 8 \frac{(4)}{K_3} \frac{(4)}{K_9}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\frac{8 \frac{(4)}{K_3}{}^2 + 8 \frac{(4)}{K_3} \frac{(4)}{K_4} + 3 \frac{(4)}{K_4}{}^2 - 8 \frac{(4)}{K_4} \frac{(4)}{K_9} + 16 \frac{(4)}{K_9}{}^2}{(\frac{(4)}{K_3} + \frac{(4)}{K_4} - 2 \frac{(4)}{K_9}) (\frac{(4)}{K_4} + 8 \frac{(4)}{K_{15}} (\frac{(4)}{K_3} + \frac{(4)}{K_4} - 2 \frac{(4)}{K_9}) + 8 \frac{(4)}{K_3} \frac{(4)}{K_9})}$                                                                            |
| Resolved unitarity condition(s):                                                   | (Demonstrably impossible)                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                  |